# MYSQL - SQL Practical

## Practical Activity

*1.- Navigate to the following website: [Programiz]*

*2.- Copy and paste the create table and insert commands, given below, into the input screen and RUN SQL:*

---

```
drop table Customers;

drop table Orders;

drop table Shippings;

 -- Create the 'employee' table

CREATE TABLE employee (

    employee_id INT PRIMARY KEY,

    employee_name VARCHAR(50),

    department_id INT,

    hire_date DATE,

    salary DECIMAL(10, 2)

);

 -- Create the 'department' table

CREATE TABLE department (

    department_id INT PRIMARY KEY,

    department_name VARCHAR(50)

);
```

```
-- Create the 'product' table
CREATE TABLE product (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(50),
    price DECIMAL(10, 2),
    stock_quantity INT
);
-- Create the 'purchase' table
CREATE TABLE purchase (
    order_id INT PRIMARY KEY,
    employee_id INT,
    product_id INT,
    order_date DATE,
    quantity INT,
    FOREIGN KEY (employee_id) REFERENCES employee(employee_id),
    FOREIGN KEY (product_id) REFERENCES product(product_id)
);
-- Insert data into the 'employee' table
INSERT INTO employee VALUES
(1, 'John Doe', 1, '2022-01-01', 50000.00),
(2, 'Jane Smith', 2, '2022-02-15', 60000.00),
(3, 'Bob Johnson', 1, '2022-03-20', 55000.00),
(4, 'Alice Brown', 4, '2022-04-10', 70000.00),
(5, 'Charlie Wilson', 2, '2022-05-05', 65000.00);
```

```
-- Insert data into the 'department' table
INSERT INTO department VALUES
(1, 'HR'),
(2, 'Finance'),
(3, 'Sales'),
(4, 'IT'),
(5, 'Marketing');
-- Insert data into the 'product' table
INSERT INTO product VALUES
(1, 'Laptop', 1200.00, 50),
(2, 'Smartphone', 800.00, 100),
(3, 'Printer', 200.00, 30),
(4, 'Desk Chair', 150.00, 20),
(5, 'Coffee Maker', 50.00, 10);
-- Insert data into the 'purchase' table
INSERT INTO purchase VALUES
(1, 1, 2, '2023-01-05', 3),
(2, 3, 1, '2023-02-10', 2),
(3, 2, 3, '2023-03-15', 1),
(4, 4, 5, '2023-04-20', 4),
(5, 5, 4, '2023-05-25', 2);
```

*You should have the 4 tables with the respective records now created.*

*You should arrive at a window that has the following:*

**Department [-]**

- department_id [int]
- department_name [varchar(50)]

**Employee [-]**

- employee_id [int]
- employee_name [varchar(50)]
- department_id [int]
- hire_date [date]
- salary [decimal(10, 2)]

**Product [-]**

- product_id [int]
- product_name [varchar(50)]
- price [decimal(10, 2)]
- stock_quantity [int]

**Purchase [-]**

- order_id [int]
- employee_id [int]
- product_id [int]
- order_date [date]
- quantity [int]

---

*3.- Solve the queries asked for, take screenshots and save them. Remember to take a copy of your SQL code as well. These should be placed in a word document with the heading "SQL Project". These are not used for assessment, but can be used as supporting evidence when applying for our boot-camp.*

---

1.- Select all columns from the employee table.

Expected Output:

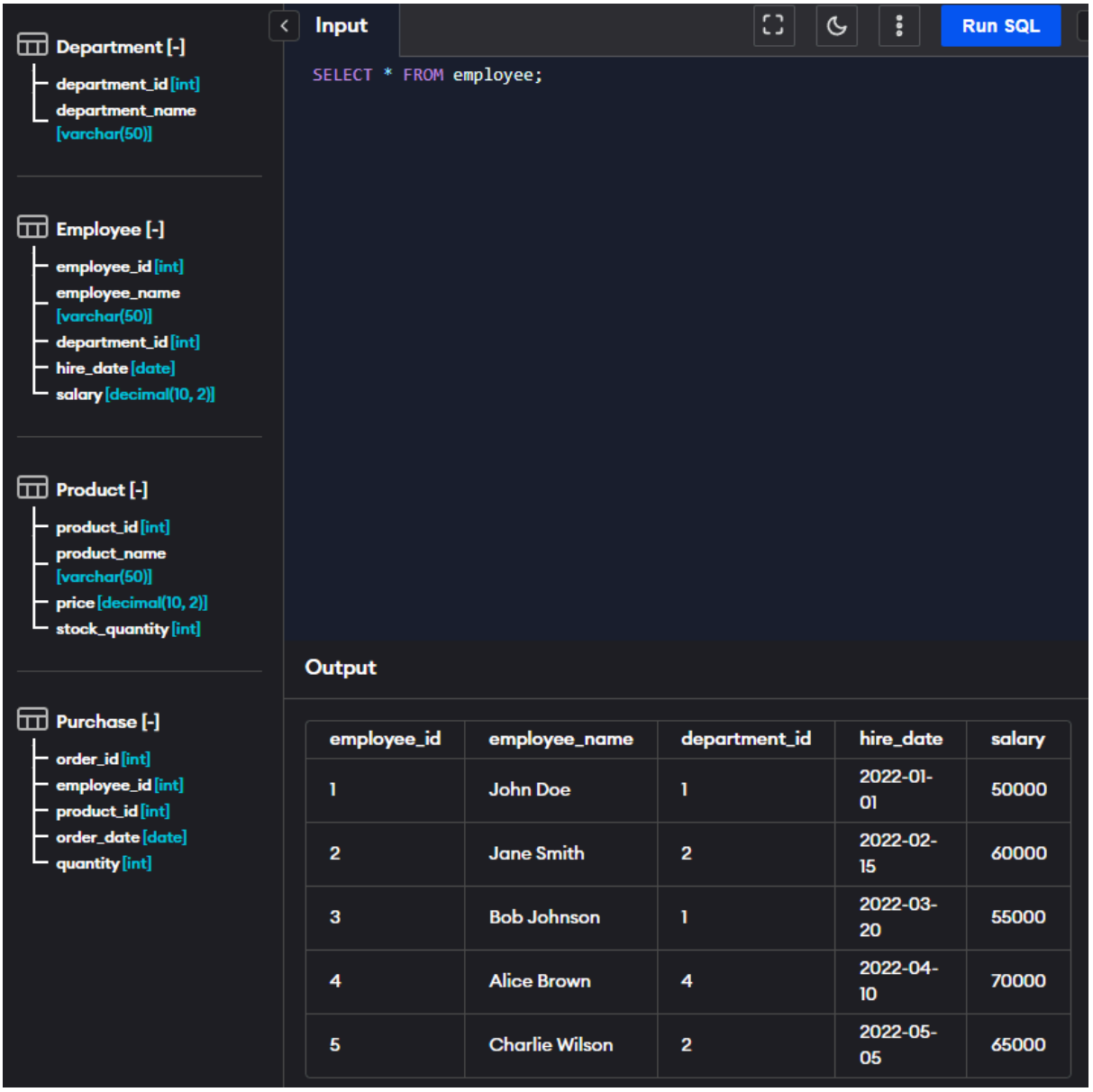
Output

| employee_id | employee_name | department_id | hire_date | salary |
|---|---|---|---|---|
| 1 | John Doe | 1 | 2022-01-01 | 50000 |
| 2 | Jane Smith | 2 | 2022-02-15 | 60000 |
| 3 | Bob Johnson | 1 | 2022-03-20 | 55000 |
| 4 | Alice Brown | 4 | 2022-04-10 | 70000 |
| 5 | Charlie Wilson | 2 | 2022-05-05 | 65000 |

*(Please see below the SQL code and the screenshot, thank you)*

SQL Code used:

SELECT * FROM employee;




| employee_id | employee_name | department_id | hire_date | salary |
|---|---|---|---|---|
| 1 | John Doe | 1 | 2022-01-01 | 50000 |
| 2 | Jane Smith | 2 | 2022-02-15 | 60000 |
| 3 | Bob Johnson | 1 | 2022-03-20 | 55000 |
| 4 | Alice Brown | 4 | 2022-04-10 | 70000 |
| 5 | Charlie Wilson | 2 | 2022-05-05 | 65000 |

2.- Select unique department names from the "department" table:

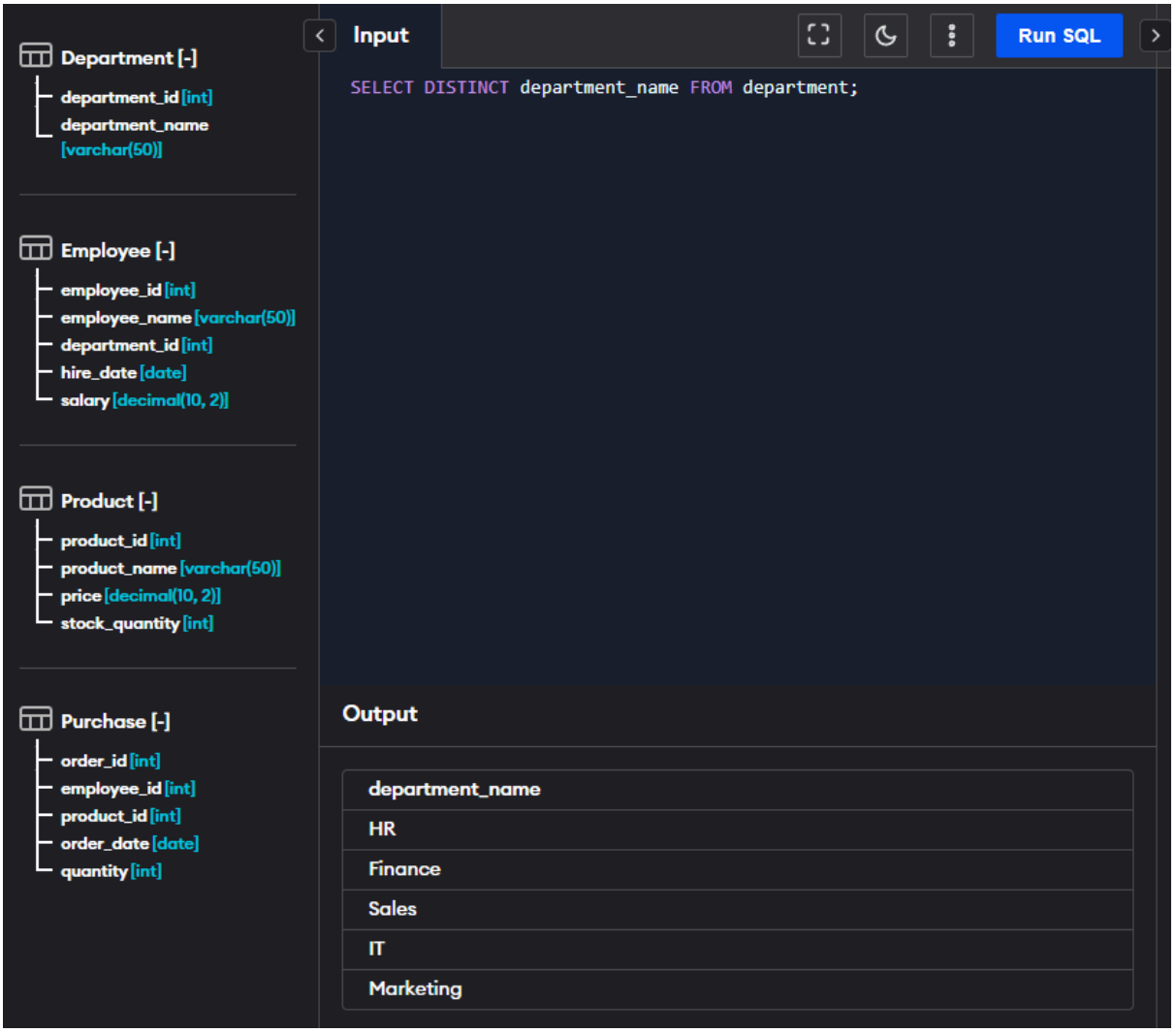
Expected Output:

**Output**

| department_name |
| --- |
| HR |
| Finance |
| Sales |
| IT |
| Marketing |

SQL Code used:

SELECT DISTINCT department_name FROM department;



3.- Select employee names and salaries from the "employee" table, ordered by salary in descending order:

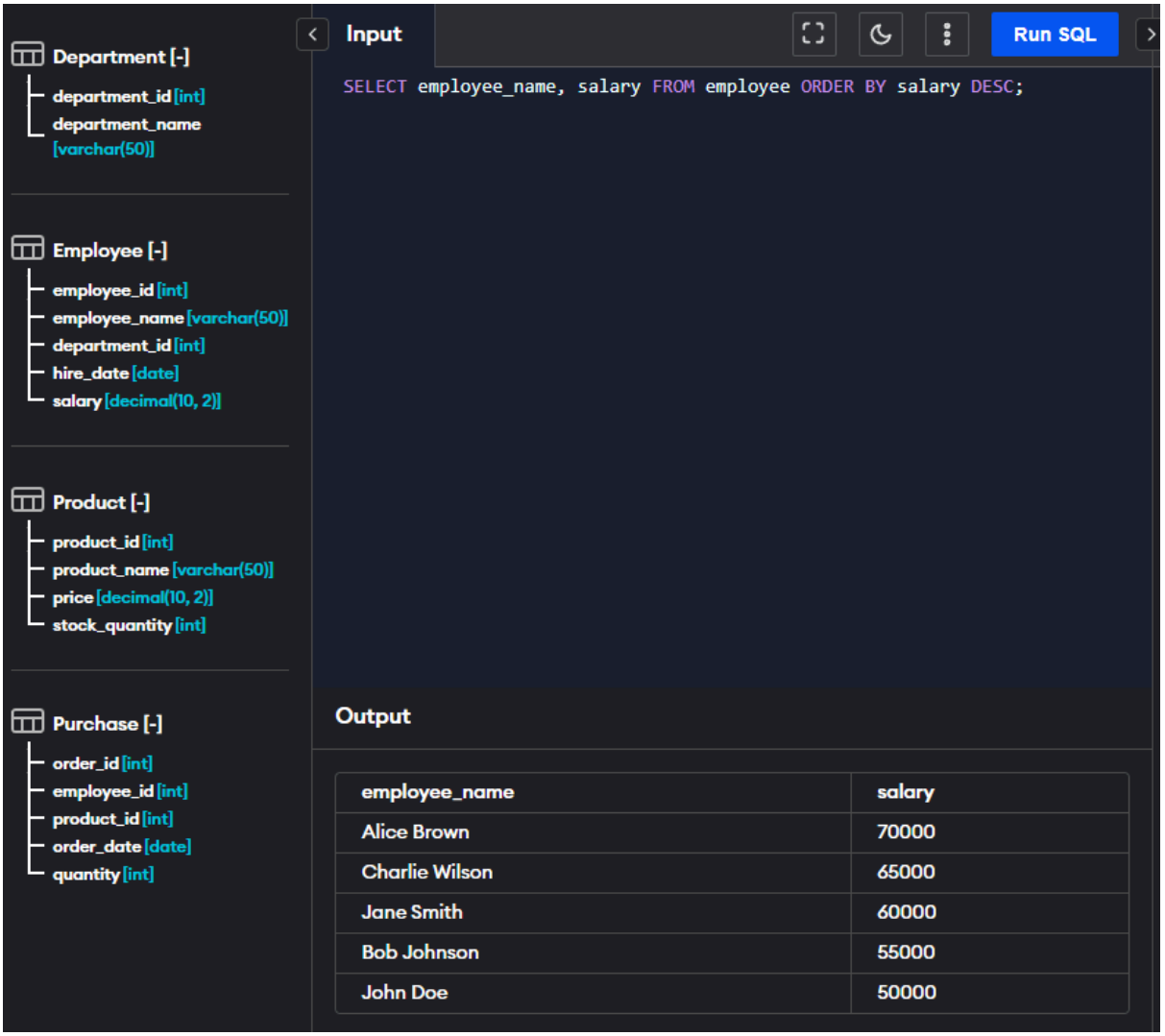
Expected Output:

**Output**

| employee_name | salary |
|---|---|
| Alice Brown | 70000 |
| Charlie Wilson | 65000 |
| Jane Smith | 60000 |
| Bob Johnson | 55000 |
| John Doe | 50000 |

SQL Code used:

SELECT employee_name, salary FROM employee ORDER BY salary DESC;



4.- Select products with a price between £100 and £500 from the "product" table:

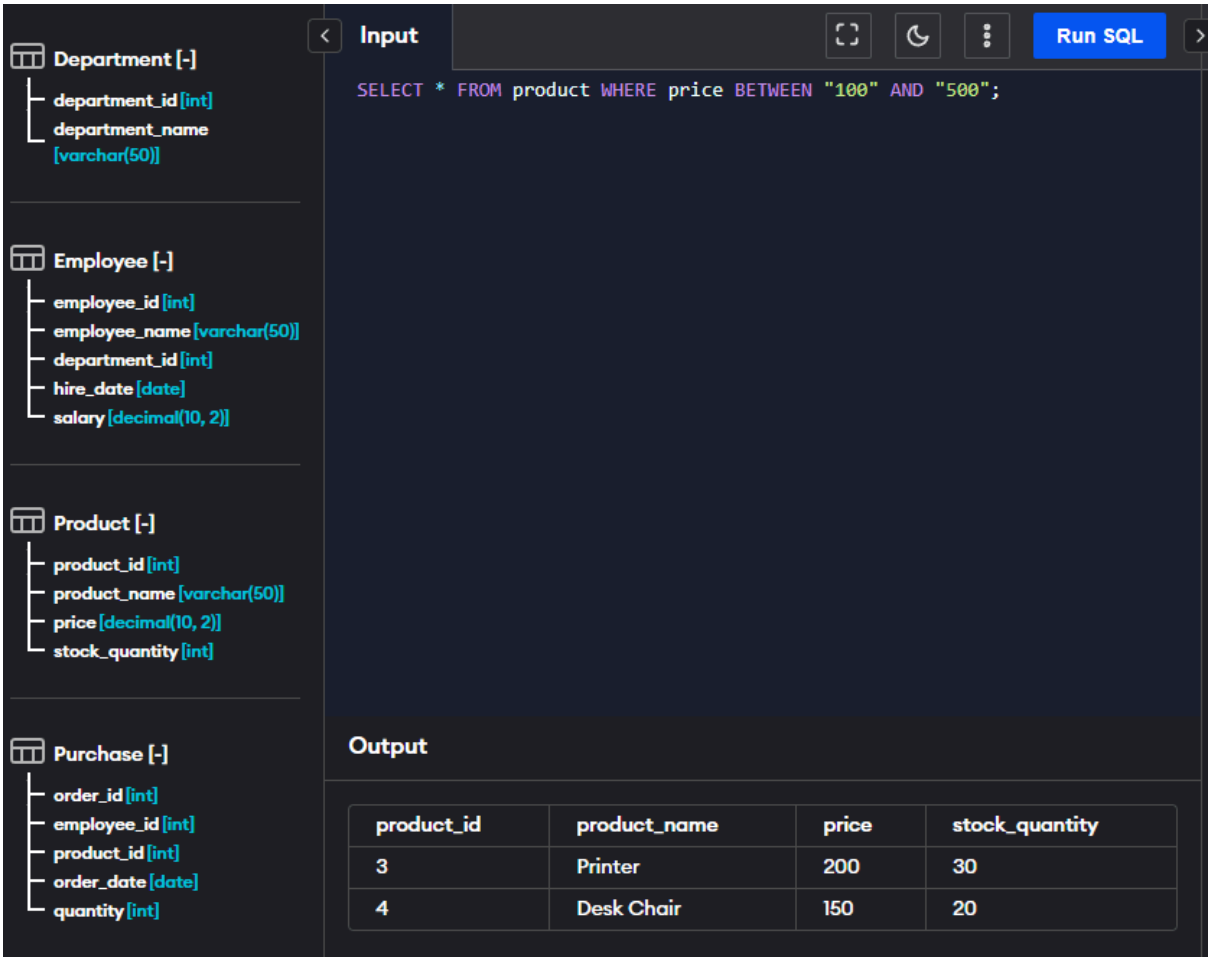
Expected Output:

**Output**

| product_id | product_name | price | stock_quantity |
|---|---|---|---|
| 3 | Printer | 200 | 30 |
| 4 | Desk Chair | 150 | 20 |

SQL Code used:

SELECT * FROM product WHERE price BETWEEN “100” AND “500”;



5.- Select orders made by employees in the "IT" department by joining "purchase" and "employees" tables:

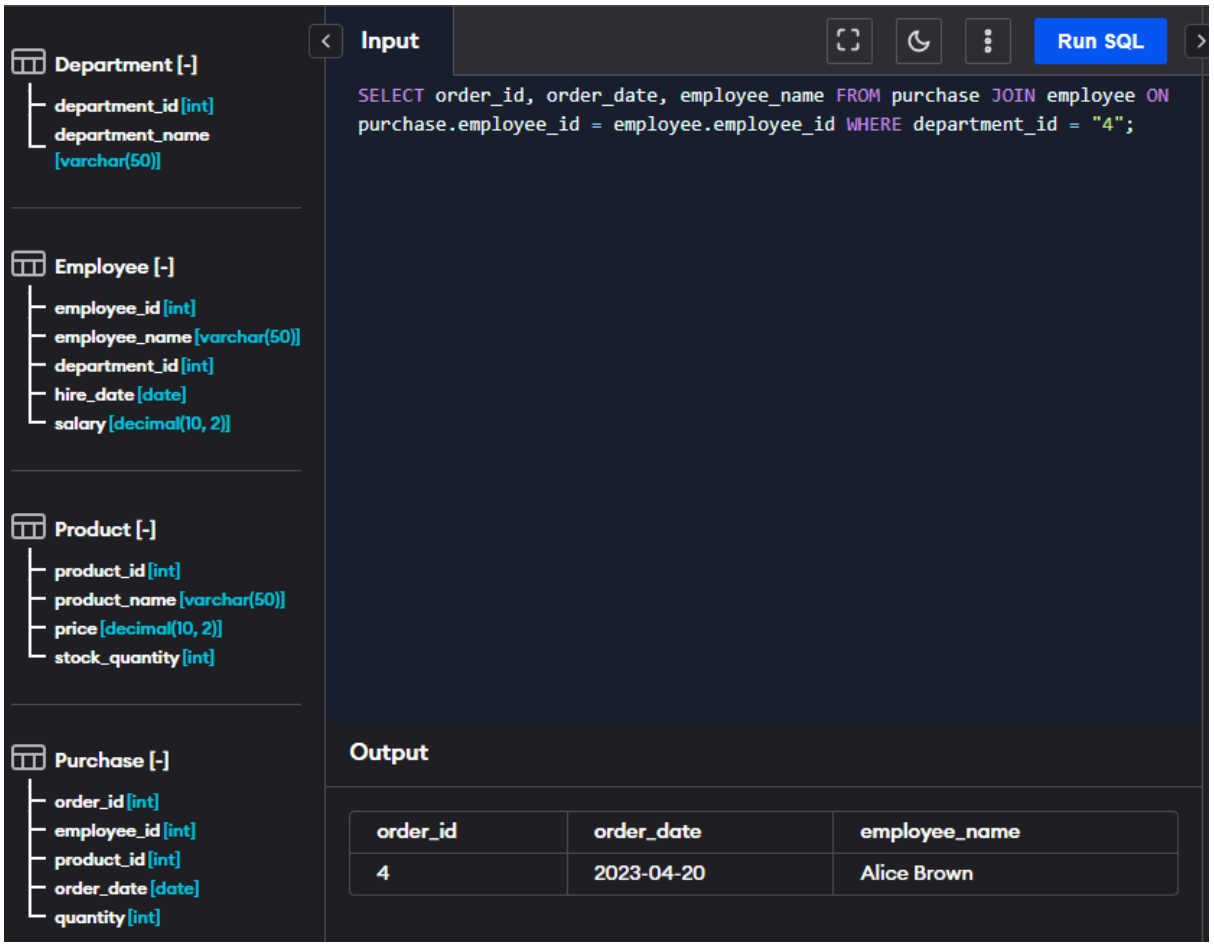
Expected Output:

**Output**

| order_id | order_date | employee_name |
|---|---|---|
| 4 | 2023-04-20 | Alice Brown |

SQL Code used:

SELECT order_id, order_date, employee_name FROM purchase JOIN employee ON purchase.employee_id = employee.employee_id WHERE department_id = "4";



6.- Select products with names containing “Chair” using a wildcard:

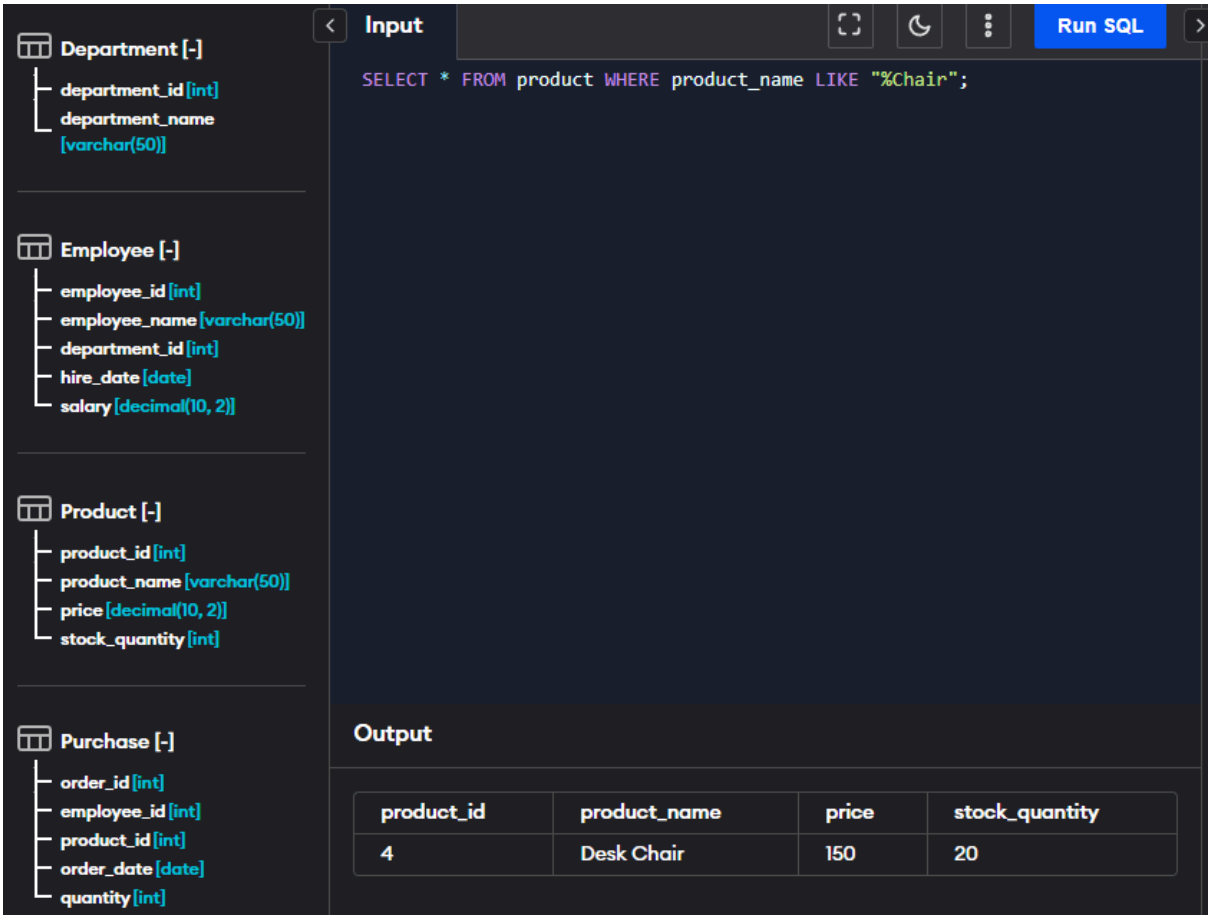
Expected Output:

Output

| product_id | product_name | price | stock_quantity |
|---|---|---|---|
| 4 | Desk Chair | 150 | 20 |

SQL Code used:

SELECT * FROM product WHERE product_name LIKE “%Chair”;



7.- Select orders made by employees with IDs 1, 3, and 5:

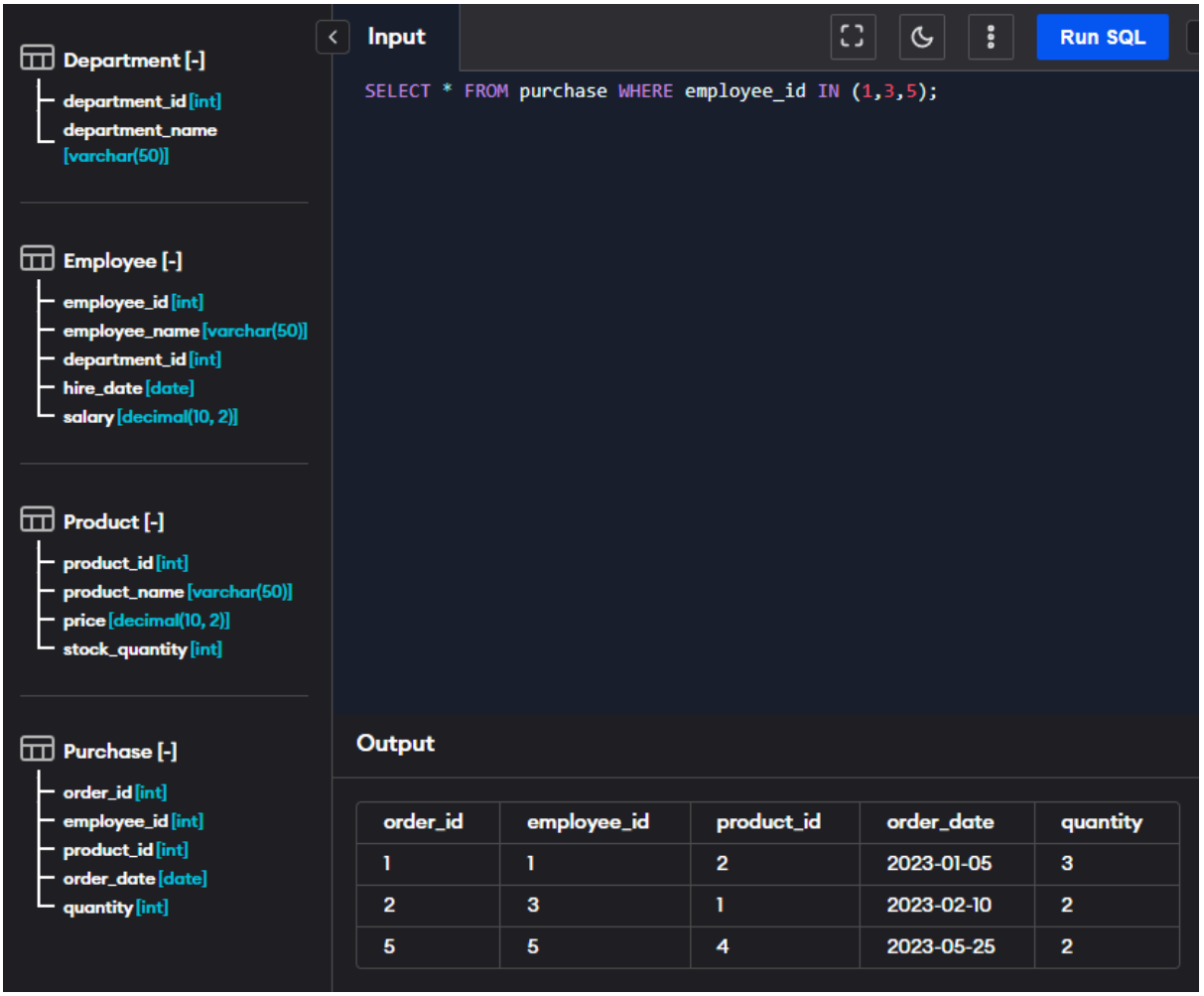
Expected Output:

Output

| order_id | employee_id | product_id | order_date | quantity |
|---|---|---|---|---|
| 1 | 1 | 2 | 2023-01-05 | 3 |
| 2 | 3 | 1 | 2023-02-10 | 2 |
| 5 | 5 | 4 | 2023-05-25 | 2 |

SQL Code used:

SELECT * FROM purchase WHERE employee_id IN (1,3,5);



8.- Select the average salary of all employees:

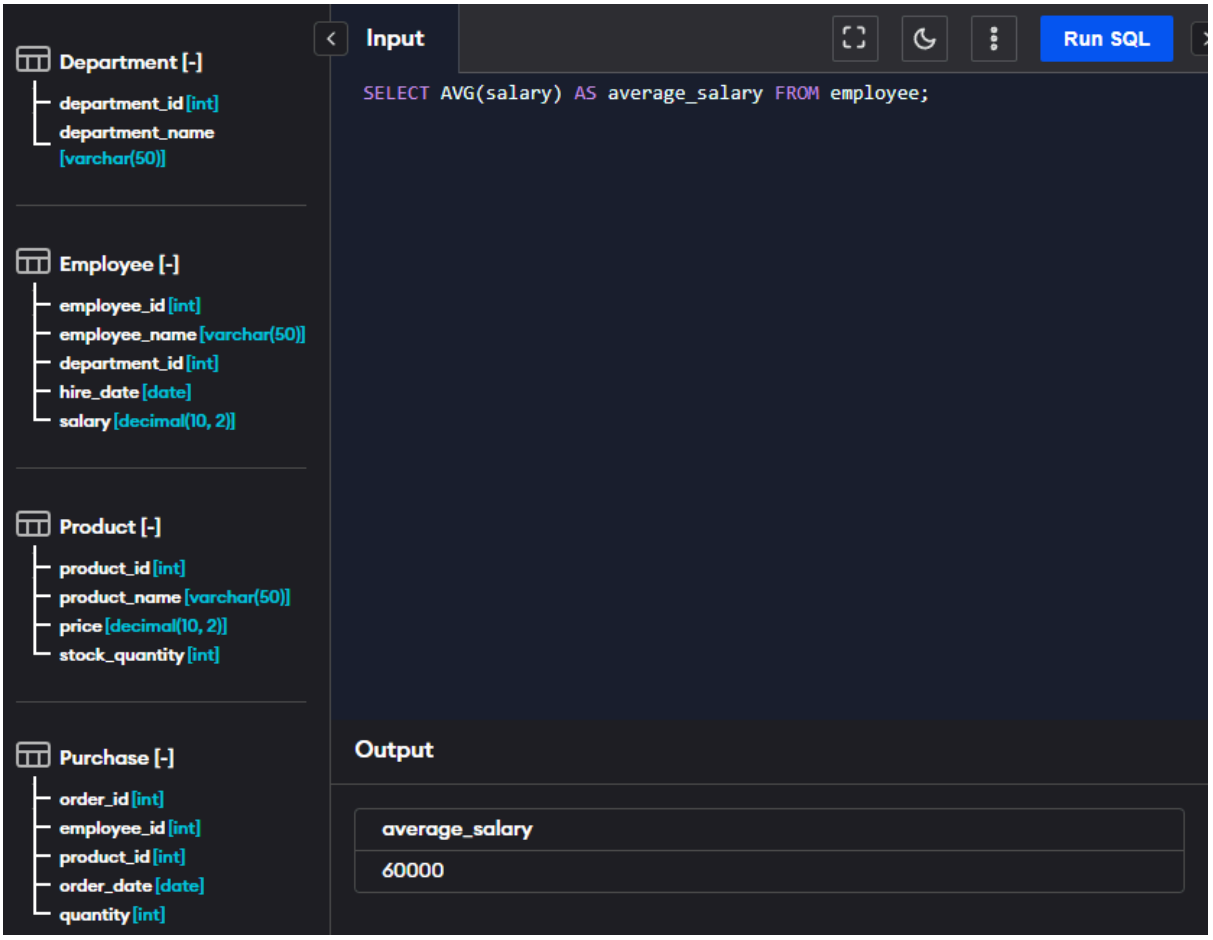
Expected Output:

Output

| average_salary |
| --- |
| 60000 |

SQL Code used:

SELECT AVG(salary) AS average_salary FROM employee;



9.- Select employees hired after "2022-03-01" with a salary greater than £60000;

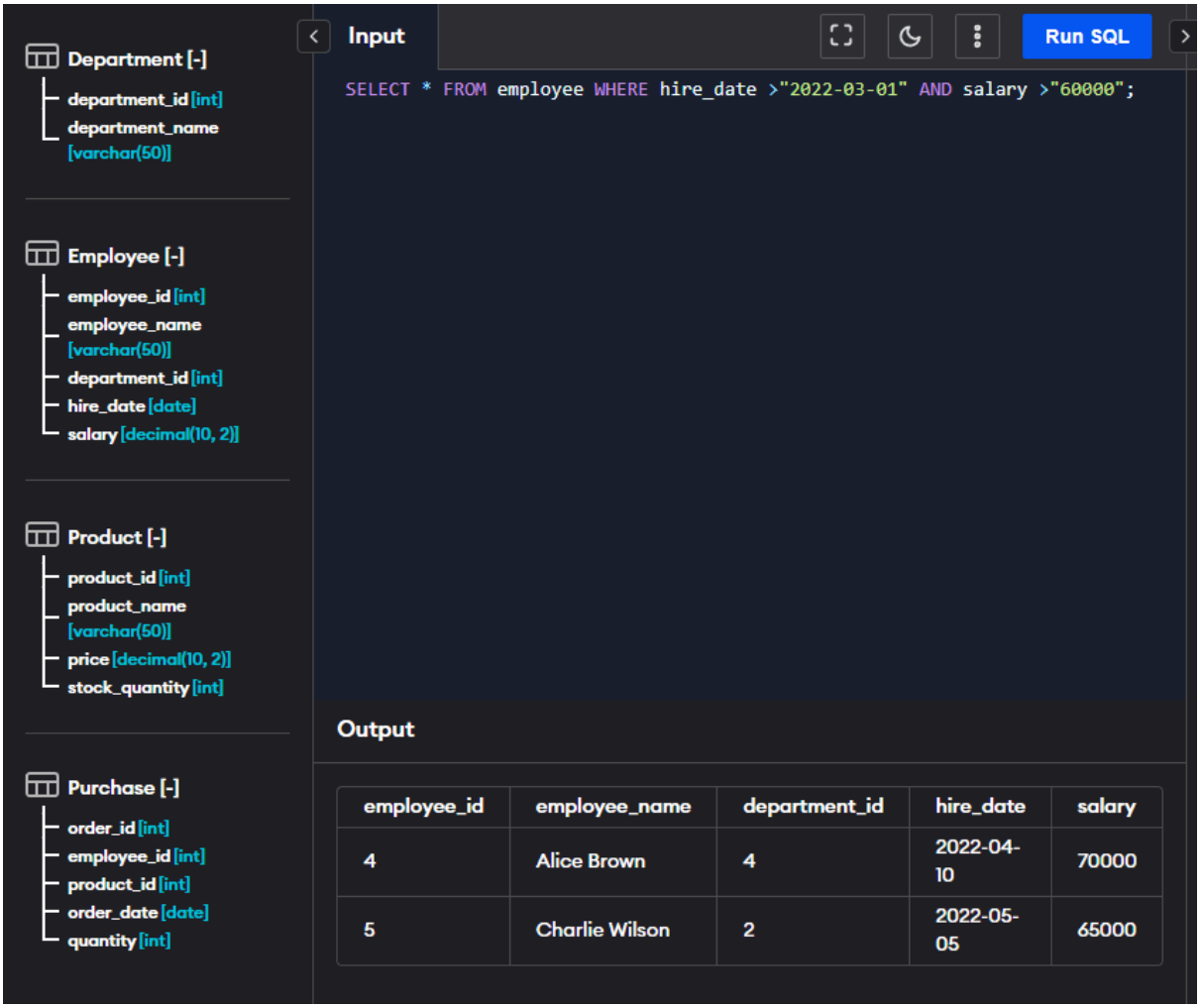
Expected Output:

Output

| employee_id | employee_name | department_id | hire_date | salary |
|---|---|---|---|---|
| 4 | Alice Brown | 4 | 2022-04-10 | 70000 |
| 5 | Charlie Wilson | 2 | 2022-05-05 | 65000 |

SQL Code used:

SELECT * FROM employee WHERE hire_date > “2022-03-01” AND salary > “60000”;

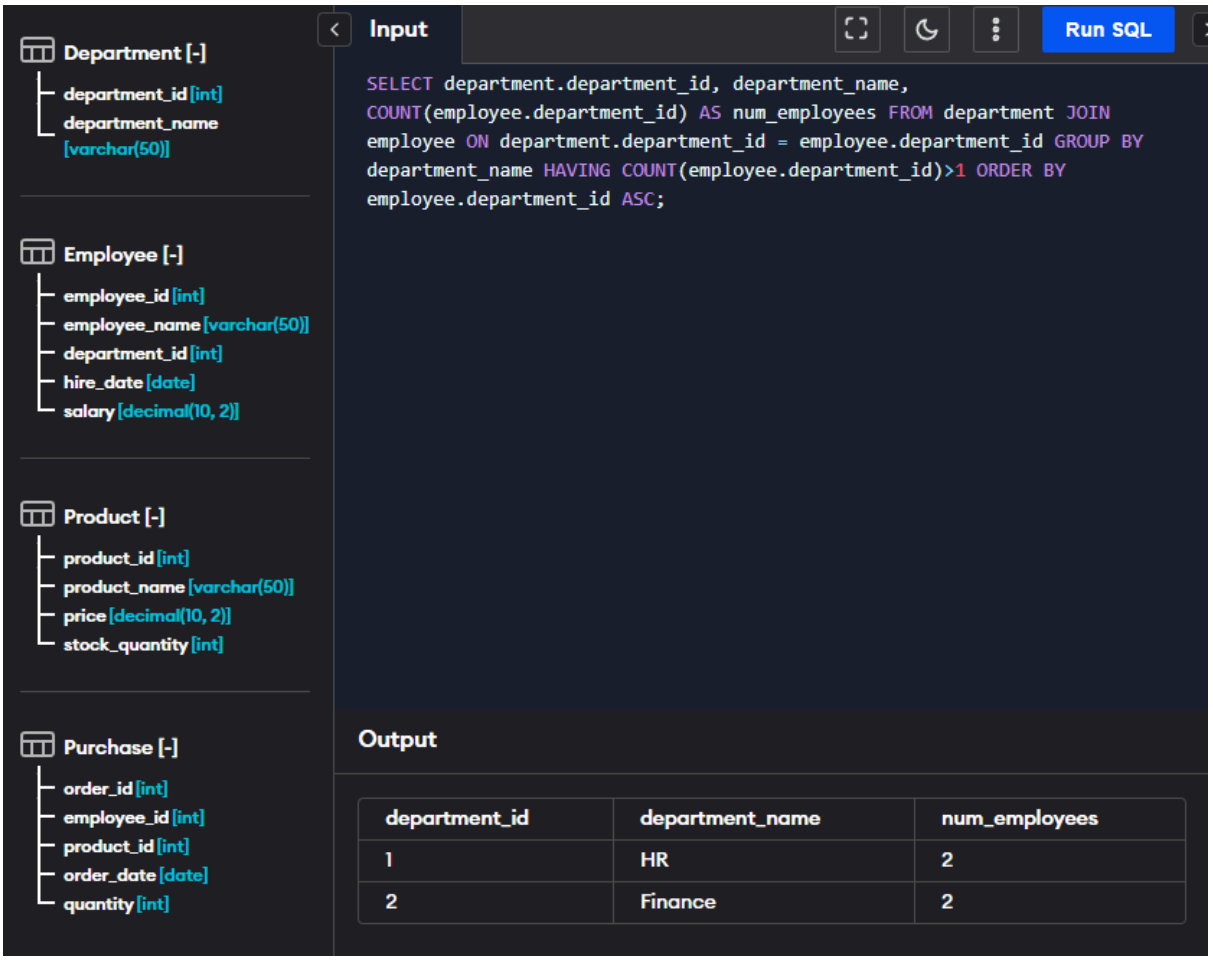


10.- Select departments with more than 1 employee, counting the number of employees in each department:

Expected Output:

**Output**

| department_id | department_name | num_employees |
|---|---|---|
| 1 | HR | 2 |
| 2 | Finance | 2 |

SQL Code used:

SELECT department.department_id, department_name, COUNT(employee.department_id) AS num_employees FROM department JOIN employee ON department.department_id = employee.department_id GROUP BY department_name HAVING COUNT (employee.department_id)>1 ORDER BY employee.department_id ASC;